Name:_____

Making Some Ionic Compounds

A. Heat some magnesium ribbon over a portable Bunsen. Don't look at the flames. Wear safety glasses.

1. What is the symbol for Magnesium? _____
2. What is the symbol for Oxygen? _____

3. What colour was the material produced?

_____

The material produced is Magnesium oxide.

When the Magnesium burns each magnesium atom loses 2 electrons and forms the ion

Mg^{+2}

The oxygen atoms in the air near the burning magnesium gain two electrons and form the ion

O^{-2}

These ions have different charges (positive and negative) so they will join together with _____ bonds.

4. Write the formula for the magnesium oxide.

B. Sprinkle some iron filings over the blue heating flame of the portable Bunsen. WEAR SAFETY GLASSES

1. What is the symbol for Iron? _____
2. What colour was the material produced?

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3. The burning causes the iron to mix with the oxygen. What is the name of the substance produced?

_____

When the iron burns each iron atom loses 3 electrons and forms the ion



The oxygen atoms in the air near the burning iron gain two electrons and form the ion



These ions have different charges (positive and negative) so the will bond together to form _____ bonds.

3. Write the formula for the iron oxide.

C. Heat some fine copper wire over a portable Bunsen. Wear safety glasses.

1. What is the symbol for Copper? _____
2. What colour was the material produced?

3. What is the name of the material produced?



Copper is a metal, so [ ] when the copper burns each copper atom _____ 2 electrons and forms the ion:

Oxygen is a _____ _____, so oxygen atoms in the air near the burning copper _____ two electrons and form the ion:

[ ]

These ions have different charges (copper is _____ and oxygen _____) so they will bond together to form _____ bonds.

4. Write the formula for the copper oxide.

Balancing Simple Chemical Equations.

Balance and write the names of the following formulas.

1. Al PO_4

   _____

2. Zn PO_4

   _____

3. Pb NO_3

   _____

4. Cu SO_4

   _____

5. Zn HCO_3

   _____

CATIONS

1+ Charge		2+ Charge		3+ Charge	
H^{1+}	hydrogen	Mg^{2+}	magnesium	Al^{3+}	aluminum
Na^{1+}	sodium	Ca^{2+}	calcium	Fe^{3+}	iron(III)
K^{1+}	potassium	Ba^{2+}	barium	Cr^{3+}	chromium(III)*
Cu^{1+}	copper(I)	Zn^{2+}	zinc		
Ag^{1+}	silver	Cd^{2+}	cadmium		
NH_4^{1+}	ammonium	Hg^{2+}	mercury(II)		
Li^{1+}	lithium	Cu^{2+}	copper(II)		
		Pb^{2+}	lead(II)		
		Fe^{2+}	iron(II)		

ANIONS

1– Charge		2– Charge		3– Charge	
F^{1-}	fluoride	O^{2-}	oxide	PO_4^{3-}	phosphate
Cl^{1-}	chloride	S^{2-}	sulfide	P^{3-}	phosphide
Br^{1-}	bromide	SO_4^{2-}	sulfate	N^{3-}	nitride
I^{1-}	iodide	SO_3^{2-}	sulfite		
NO_3^{1-}	nitrate	CO_3^{2-}	carbonate		
NO_2^{1-}	nitrite				
OH^{1-}	hydroxide				
CN^{1-}	cyanide				
$CH_3CO_2^{1-}$	acetate				
HCO_3^{1-}	bicarbonate				
ClO^{1-}	hypochlorite				
SCN^{1-}	thiocyanate				
ClO_3^{1-}	chlorate				